

F1 Track Limits Digital Scrutineering System Blueprint

1. The Core Approach

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- **The Problem:** Low-frequency telemetry transponders (~3–4 Hz GPS feeds in public tools like FastF1) suffer from noise and geometric drift, meaning coordinates alone cannot legally prove a sub-inch tire-to-line violation.
- **The Strategy:** A multi-modal pipeline combining **deterministic spatial filtering** (Shapely + TUMFTM) to catch candidates, a **high-frequency tabular machine learning classifier** (LightGBM) to isolate driver input anomalies, and **zero-shot computer vision masking** (Grounding DINO + Meta SAM 2) for pixel-level verification.
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2. Fully Loaded Tech Stack

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- **Ingestion & Telemetry:** FastF1 API (Lap telemetry profiles: Speed, Throttle, Brake, Steering, RPM, G-Forces).
- **Geospatial Processing:** Shapely + TUMFTM Racetrack Database (Building outer shells and inner cutout hole polygons).
- **Dashboard Frontend:** Streamlit or Gradio for rapid presentation UI, filtering panels, and evidence media playback.
- **Tabular AI Analytics:** LightGBM (Optimized gradient boosting framework for time-series row-window sequences).
- **Computer Vision Pipeline:** Grounding DINO (Zero-shot text-prompted bounding boxes) paired with Meta SAM 2 (High-fidelity pixel segmentation masking).
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The 3-Step Implementation Strategy

[FastF1 Telemetry Feed] ---> [Min-Max Alignment] ---> [Step 1: Shapely Polygon Check] ---> Map (Red Alerts)

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Generates 20-Row Window Windows

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[Broadcast Evidence Video] -> [Grounding DINO] -----> [Step 2: LightGBM Tabular Classifier] -> Probability Score

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[Synced UI Evidence Room] <--- [Meta SAM 2] <----- [Phase 3: Integrated Video & AI Verification]

Phase 1: Spatial Ground Truth Engine (Deterministic MVP)

1. **Extract Targets:** Query session records via FastF1 and capture officially deleted laps (`laps[laps['Deleted'] == True]`) to use as historical ground truth.
2. **Coordinate Calibration:** Map arbitrary telemetry frames to absolute surveying coordinates from the TUMFTM repository using min-max scaling to compensate for global positional drifts.
3. **Polygon Containment:** Initialize a 2D track layout polygon in Shapely. Execute continuous checks using `not track_polygon.contains(Point(x, y))` to immediately locate points breaching the white track lines and render them as **red alert dots** over your dashboard track layout.

Phase 2: Telemetry Dataset Engineering & LightGBM Anomaly Modeling

1. **Building the Dataset:** For every corner apex across all laps, extract a synchronized **20-row sliding time-series window** containing the raw input arrays: [Speed, Throttle, Brake, Steering, RPM, G-Force_Lat, G-Force_Long]. Assign a binary target label: **1 (Violation)** if flagged as an official infraction or caught by the spatial bounds check, and **0 (Legal Line)** if compliant.
2. **The LightGBM Advantage:** Coordinates can drift, but *driver inputs do not lie*. Exceeding track boundaries forces distinctive vehicle physics—such as rapid throttle lift-offs, high-frequency steering micro-corrections over exit kerbs, and abrupt lateral G-force drops. **LightGBM** is selected as the top architecture choice because it processes complex multi-variable tabular rows significantly faster than neural networks during high-pressure hackathons. The model outputs a continuous *Anomaly Probability Score* to prioritize the stewards' review queue.

Phase 3: GenAI Labeling Pipeline & Video Evidence Integration

1. **Evidence Syncing:** Stage short 5–10s clips from official feeds inside a local project folder indexed as `clips/{DRIVER}_lap_{LAP_NUM}.mp4`. Clicking an infraction row inside the Streamlit table dynamically boots the video next to the telemetry charts.
2. **GenAI / SAM 2 Automation Horizon:** For automated labeling and validation, route video frames through a zero-shot object detector (Grounding DINO) with a text prompt like "Formula 1 tyre" or "white track line" to isolate regions of interest. Pass these bounding boxes directly to **Meta SAM 2** to generate pixel-perfect masks of the tire profile against the painted edge line. Intersections between masks across frames provide a robust verification pipeline for your pitch slides.

Hackathon Pitch Structure Checklist

1. **The Hook (1 Min):** Show the operational strain on the FIA where stewards spend hours reviewing lines post-race.
2. **The Demo (2 Mins):** Open the live **Streamlit Dashboard**. Filter a driver, display the aligned **TUMFTM map layout**, show the **red alert markers**, and play the synced video asset.

3. **The Deep Tech (1 Min):** Explain the **LightGBM behavioral model** (detecting driver corrections over kerbs) and show the **SAM 2 pixel-masking framework** on your presentation slides.
4. **The Business Case (1 Min):** Pitch this as a proactive *Stewards' Decision Assistant* built for immediate production scalability.